to unsafe BeaverTails samples [29]. Conversely, for emergent misalignment, we first fine-tune on misaligned response data [22] before reverting to benign response data. This setup allows us to observe the Rebound Force $\mathcal{F}_{\text{rebound}}$ in both "aligned" and "misaligned" trajectories.

Experiment Results. As shown in Figure 2, the model demonstrates a rapid rebound to the baseline, a phenomenon that remains invariant to the duration of forward fine-tuning in Stage 1. This is consistent with our theory: the **Rebound Force** $\mathcal{F}_{\text{rebound}}$ resists forward fine-tuning (Stage 1) and facilitates reverse fine-tuning (Stage 2). This pattern holds across diverse models and datasets (Figure 5). Moreover, our findings reveal a dual "instability": just as aligned states are "unstable", misaligned states are also "unstable". Following fine-tuning on a misaligned dataset, the model's misalignment rate drops precipitously when retrained on benign data (Figure 2(b)). Remarkably, only a few SFT steps are required to effectively mitigate these misaligned behaviors.

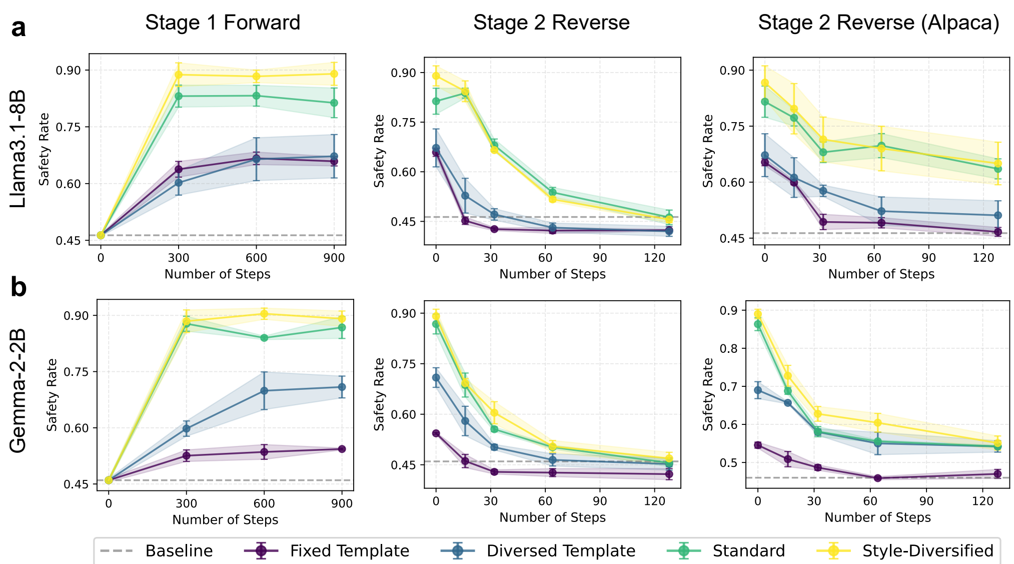

Figure 3: **Impact of distribution narrowness on rebound force**. Results are shown for (a) Llama3.1-8B and (b) Gemma-2-2b. "Stage 2 Reverse (Alpaca)" denotes the application of the alpaca-cleaned dataset for stage 2 reverse fine-tuning. The narrowness of dataset distribution increases across four levels: *Fixed Template* (highest), *Diversified Template*, *Standard*, and *Style-Diversified* (lowest). Lighter colors indicate higher dataset diversity (lower distribution narrowness).

3.3 The Role of Distribution Narrowness in Rebound Dynamics

Motivation. Our dynamical framework (Theorem 2.9) yields a specific, testable prediction: the magnitude of the **Rebound Force** $\mathcal{F}_{\text{rebound}}$ is directly modulated by the term $\tilde{\boldsymbol{\pi}}^{+\top}\boldsymbol{K}\tilde{\boldsymbol{\pi}}^{+}$. This term captures the narrowness of the model's aligned posterior distribution $\tilde{\boldsymbol{\pi}}^{+}$.

Experimental Configurations. To verify this, we construct a narrowness gradient using four safety-alignment datasets with decreasing diversity: *Fixed Template* , *Diversified Template*, *Standard*, and *Style-Diversified* (ordered by decreasing narrowness; see Appendix B.3.2 for construction details). We then execute the multi-stage SFT paradigm with two Stage 2 variations. **1) Standard Reverse:** Fine-tuning on unsafe HEx-PHI completions [8] to observe typical rebound. **2) Task-Agnostic Reverse:** Fine-tuning on the Alpaca-cleaned dataset [28]. Since this dataset is unrelated to safety, it minimizes the task-specific external $\mathcal{F}_{\text{drive}}$, thereby isolating the effect of $\mathcal{F}_{\text{rebound}}$ on the safety rate.

Experiment Results. As shown in Figure 3, dataset diversity systematically affects both alignment and reversal. In Stage 1, models trained on more diverse datasets achieve higher safety scores, while narrower datasets lead to lower convergence points, consistent with stronger rebound under narrower distributions. In Stage 2, models trained on narrower datasets exhibit faster safety degradation at matched safety rate levels. This trend persists under both standard and task-agnostic (Alpaca) reversal, indicating that rebound dynamics are governed by distribution narrowness.

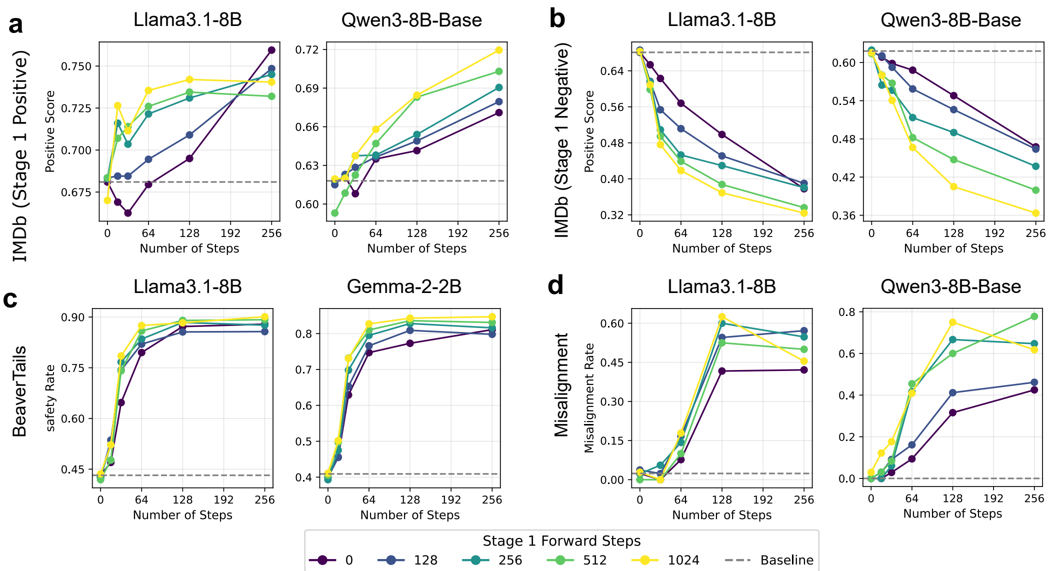

Figure 4: **Dynamics of alignment score S during Stage 3 (Re-exposure)**. Subplots (a) and (b) present results on the IMDb dataset, initialized with Stage 1 fine-tuning on positive and negative sentiments, respectively. Subplots (c) and (d) show results for BeaverTails and Emergent Misalignment. The color gradient represents the duration of Stage 1 fine-tuning, where lighter lines denote a higher number of training steps. The results demonstrate that prolonged Stage 1 fine-tuning significantly accelerates the fine-tuning speed in Stage 3.

4 The Rehearsal Priming Effect: Re-Exposure Accelerates Alignment

Rehearsal Priming Effect. We introduce **Rehearsal Priming**, a dynamical phenomenon in which an LLM adapts faster when re-exposed to a distribution encountered in a prior fine-tuning stage. Our framework attributes this effect to the residue in the posterior distributions $\tilde{\pi}^+$ and $\tilde{\pi}^-$ induced during Stage 1. While a reverse phase may suppress the alignment score S, this posterior residue can persist and, according to Theorem 2.9, directly modulate the **Driving Force** $\mathcal{F}_{\text{drive}}$. As a result, re-exposure to the same distribution yields a stronger effective driving signal, leading to faster re-alignment. This gives a clear, testable prediction: deeper initial fine-tuning (Stage 1) leads to faster adaptation upon re-exposure (Stage 3).

While prior work on re-alignment [22, 30] observed rapid recovery in safety tasks, it did not identify this systematic dependence on initial fine-tuning depth. Our dynamic framework provides a mechanistic explanation of this effect, which we systematically validate across safety, misalignment, and sentiment settings.

Experimental Protocol. To isolate the Rehearsal Priming effect, we implement a three-stage sequence: Stage 1 (Forward), Stage 2 (Reverse), and Stage 3 (Re-exposure). The primary independent variable is the exposure depth in Stage 1, varied by the number of SFT steps.

To ensure a rigorous comparison of adaptation velocities, we employ a **score-matching protocol**: Stage 3 is initialized using Stage 2 checkpoints whose alignment scores S most closely approximate the pre-trained baseline (See Appendix B.2.2 for details). This control ensures that any observed acceleration in Stage 3 stems from the latent structural residue rather than discrepancies in the starting state. We evaluate this protocol across four scenarios to verify its generality: BeaverTails [29], emergent misalignment [22], and bidirectional IMDb [31] sentiment shifts (alternating between positive and negative distributions). All models and metrics follow the Section 3.2.

Experiment Results. Figure 4 shows that deeper Stage 1 exposure leads to faster adaptation in Stage 3. This is consistent with Rehearsal Priming. The same trend holds across all datasets and model architectures, which is aligned with our theoretical prediction of **Rehearsal Priming**. Extended results in Figure 6 further confirm these identical patterns across additional models.

Of particular concern are the implications for emergent misalignment—a phenomenon where narrowly scoped fine-tuning unexpectedly induces broad misaligned behaviors that extend far beyond the target task [10, 22]. Our results demonstrate that for models exhibiting emergent misalignment, even after being realigned through benign fine-tuning, the broad misaligned behaviors can be rapidly reactivated. In such cases, only a few steps of malicious fine-tuning are sufficient to trigger a full re-emergence of the misaligned state, highlighting the inherent fragility of subsequent alignment.

5 Related Work

Linearity in LLM Fine-Tuning. Previous studies show that model behaviors [32, 33, 34], representations [35, 36], and logits [37] evolve approximately linearly during fine-tuning, including RL-based methods like RLVR [38]. Leveraging this, [18] uses empirical NTK to decompose log-probability changes into token-level contributions, providing a theoretical starting point for our analysis.

Alignment Paradigms. Safety alignment typically relies on Supervised Fine-Tuning (SFT) to internalize post-training [1, 2], followed by preference-based optimization such as RLHF [1, 3, 4], DPO [5], or RLAIF [6, 7]. While SFT is central to alignment, it also constitutes a key attack surface for harmful fine-tuning [39]. We therefore focus on the SFT regime, where both our experiments and Theorem 2.9 characterize its alignment dynamics, leaving extensions to other paradigms for future work, *noting that several intermediate results (Theorem 2.7 & lemma 2.8) are not specific to SFT and may generalize to other alignment paradigms.*

Superficial Alignment Hypothesis. The superficial alignment hypothesis, introduced by LIMA [40], posits that alignment primarily alters output style rather than core capabilities. Consistent with this, post-alignment changes concentrate in early tokens [19] and stylistic marker tokens [41]. Our Theorem 2.7 provides a mechanistic account of how such localized shifts can induce global behavioral changes. This perspective is further supported by the Superficial Safety Alignment Hypothesis (SSAH) [42], which attributes safety gains to small reorientations of decision directions.

Fragility of Alignment. LLM alignment remains vulnerable to adversarial jailbreak [43, 44], backdoor poisoning [45], and malicious fine-tuning [8, 19]. Our work primarily focuses on the mechanisms behind alignment degradation during malicious fine-tuning.

Mechanistic Perspectives on Alignment Fragility. Beyond the gradient-based and distribution-based views in the introduction, recent mechanistic interpretability studies suggest that safety alignment may rely on low-dimensional directions [46, 47] or localized circuits [12, 22, 48, 49]. How these structures relate to our eNTK-based framework remains unclear and is left for future work.

Connections to Prior Empirical Findings. Ji et al. [14] observe and term the "Rebound Effect". In Section 3.3, we show that narrower fine-tuning distributions strengthen the rebound force, consistent with Xie et al. [9], who observe that overfitting on narrow distributions can induce large aligned gradients and catastrophic forgetting of safety alignment even under benign data. While prior work shows models can be rapidly restored from malicious fine-tuning [22, 30], we generalize this pattern in Sec. 4 through our proposed Rehearsal Priming Effect and provide a theoretical account.

6 Conclusion

In this work, we presented a unified dynamical framework for understanding alignment evolution during fine-tuning. By decomposing alignment score changes into a **Rebound Force** and a **Driving Force**, our theory connects token-level optimization dynamics with distributional shifts in model's alignment behavior. This perspective not only explains the fragility of safety alignment, but also revisits the Rebound Effect through a dynamical lens, reveals a novel dependence on distribution narrowness. Building on this, we further predict the **Rehearsal Priming Effect**, which we validate across safety, misalignment, and sentiment settings. More broadly, our results suggest that alignment is governed not only by the training signal itself, but also by the posterior structure it induces, offering a principled foundation for analyzing and improving the robustness of post-training alignment.

Limitations and Future Directions. Our analysis assumes a relatively stable eNTK, whose validity at industrial scales remains unclear. Moreover, while our main results focus on SFT, Theorem 2.7 and Lemma 2.8 provide initial insight into more general settings; extending the framework to DPO and other RLHF methods remains an important direction for future work.